



PROGRAMMING 732 ASSIGNMENT

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ARTICULATION DIPLOMA 3 YEAR TO BSC
402306532

PROGRAMMING 732 ASSIGNMENT



Name and Surname: _____

Student ITS No: _____

Qualification: _____ Year of Study: _____ Semester: _____

Assignment due date: _____ Date submitted: _____

QUESTION	EXAMINER MARKS	MODERATOR MARKS	REMARKS

ASSIGNMENT INSTRUCTIONS

Please tick each box to confirm completion.

Use Times New Roman font, size 12, with 1.5 line spacing throughout the document. Apply Harvard Referencing Style for all citations and references.

For essay-style assignments, please include the following sections:

Table of Contents
Introduction
Main Body (with relevant subheadings)
Conclusion
References

Submit the assignment in PDF format on Moodle.
Use the specified cover page provided.
Include a signed declaration of originality.

DECLARATION OF ORIGINALITY:

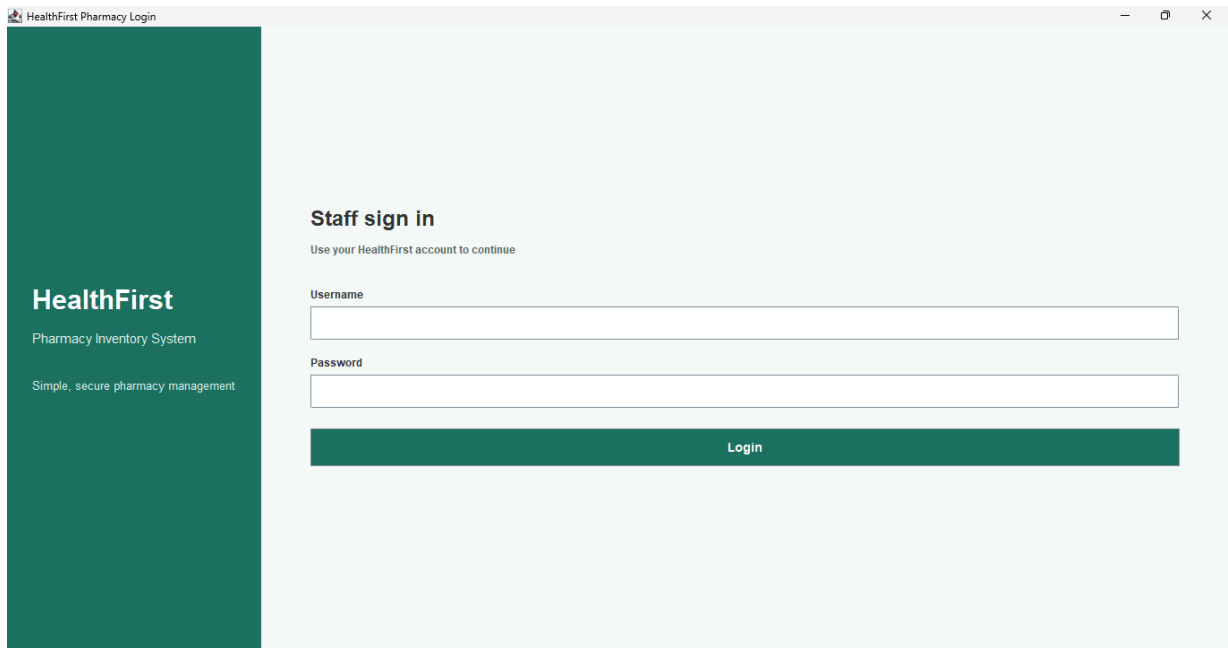
I hereby declare that this assignment is my own work and has not been copied from any other source except where due acknowledgment is made. I affirm that all sources used have been properly cited and that this submission complies with the institution's policies on academic integrity and plagiarism.

Student Signature: _____

Date: _____

Login Screen

The login screen allows Admin and Cashier users to securely access the system using their username and password. (Admin and Cashier username and password are provided in the read me file)



HealthFirst Pharmacy Login

HealthFirst
Pharmacy Inventory System
Simple, secure pharmacy management

Staff sign in
Use your HealthFirst account to continue

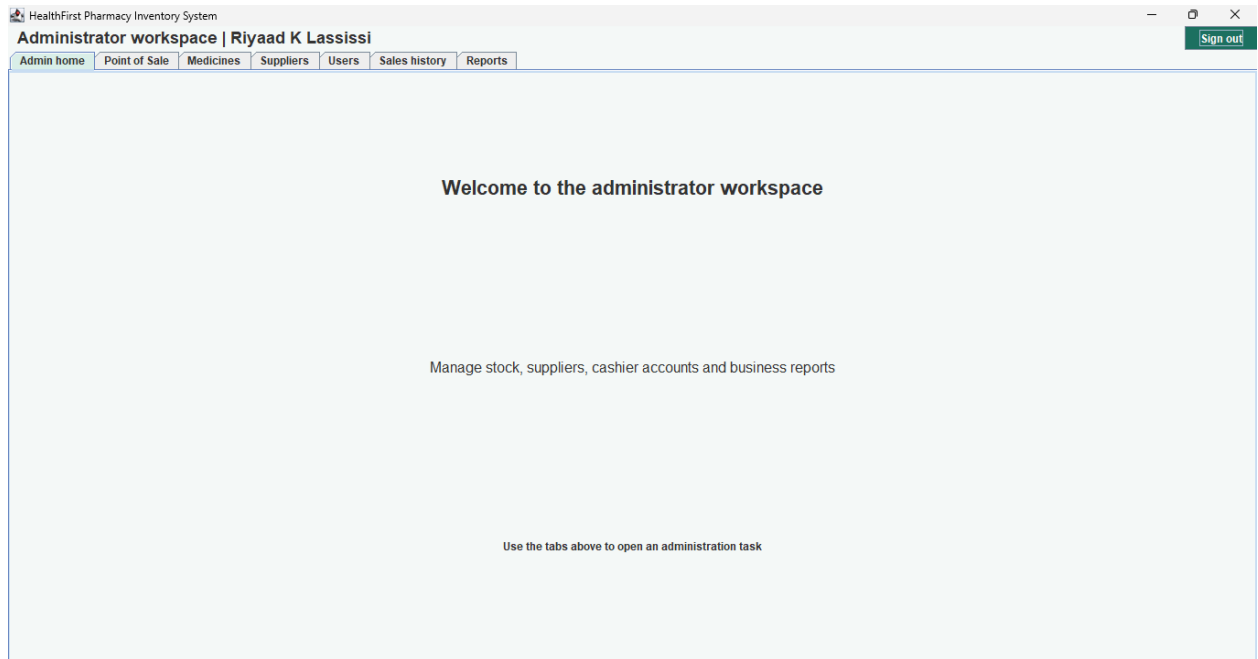
Username

Password

Login

Admin Dashboard

The Admin Dashboard displays tabs for Point of Sale, Medicines, Suppliers, Users, Sales History and Reports.



Manage Medicines tab

This tab displays medicine details such as name, company, type, category, price, stock quantity, reorder level, expiry date and supplier.

HealthFirst Pharmacy Inventory System									
Administrator workspace Riyaad K Lassissi									
Admin home Point of Sale Medicines Suppliers Users Sales history Reports Sign out									
Refresh Add medicine Edit selected Delete selected									
ID	Name	Company	Type	Category	Price	Quantity	Reorder	Expiry	Supplier
1	Paracetamol 500mg	HealthCare	Tablet	Prescribed	35.5	100	20	2027-06-30	1
2	Cough Syrup	CareChem	Syrup	Over the Counter	48.0	35	10	2026-11-30	2
3	Vitamin C	HealthCare	Tablet	Over the Counter	62.75	8	15	2026-09-15	1
4	Panado	Adcock Ingram	Tablets	Over the Counter	35.5	95	20	2027-06-30	3
5	Panado	Adcock Ingram	Syrup	Over the Counter	42.0	80	15	2027-06-30	3
6	Adcodol	Adcock Ingram	Tablets	Over the Counter	38.0	75	15	2027-06-30	3
7	Myprodol	Adcock Ingram	Capsules	Over the Counter	55.0	65	15	2027-06-30	3
8	Allergex	Adcock Ingram	Tablets	Over the Counter	45.0	70	15	2027-06-30	3
9	Allergex	Adcock Ingram	Syrup	Over the Counter	49.0	60	15	2027-06-30	3
10	Compral	Adcock Ingram	Tablets	Over the Counter	32.0	55	10	2027-06-30	3
11	Corenza C	Adcock Ingram	Tablets	Over the Counter	48.0	60	15	2027-06-30	3
12	Sinutab	Adcock Ingram	Tablets	Over the Counter	52.0	51	10	2027-06-30	3
13	Adco-Dol	Adcock Ingram	Tablets	Over the Counter	40.0	75	15	2027-06-30	3
14	Adco-Salbutamol	Adcock Ingram	Tablets	Over the Counter	46.0	50	10	2027-06-30	3
15	Adco-Zolpidem	Adcock Ingram	Tablets	Over the Counter	68.0	42	10	2027-06-30	3
16	Adco-Naproxen	Adcock Ingram	Tablets	Over the Counter	44.0	60	15	2027-06-30	3
17	Benylin	Johnson & Johnson	Syrup	Over the Counter	65.0	38	10	2027-06-30	4
18	Lindagon	Cipla	Capsules	Over the Counter	58.0	50	10	2027-06-30	5
19	Lindagon	Cipla	Syrup	Over the Counter	62.0	45	10	2027-06-30	5
20	Betadine	Kenvue	Solution	Over the Counter	72.0	35	8	2027-06-30	6
21	Strepsils	Reckitt	Lozenges	Over the Counter	39.0	84	20	2027-06-30	7

Cashier POS screen with items in a cart

The POS screen allows the cashier to add medicines to a cart, select quantities, calculate tax and complete a sale.

HealthFirst Pharmacy Inventory System

Cashier workspace | Palesa N

Point of Sale

Stock Check

Sales history

Medicine ID

13

Quantity

2

Add to cart

Clear cart

Checkout

Total: R551.20

Medicine	Type	Category	Price	Quantity	Tax	Total
12 Sinutab	Tablets	Over the Counter	52.0	1	7.8	59.8
75 Adco-Naproxen	Tablets	Over the Counter	44.0	3	19.8	151.8
42 Amoxil	Capsules	Over the Counter	82.0	2	24.599999999999998	188.6
1 Paracetamol 500mg	Tablet	Prescribed	35.5	2	0.0	71.0
13 Adco-Dol	Tablets	Prescribed	40.0	2	0.0	80.0

As South African Law indicates, Medicines that are prescribed can not be taxed but medicines bought over the counter are taxed, so admin and cashier have the option to categorise medicine on whether it has been prescribed or not.

HealthFirst Pharmacy Inventory System

Cashier workspace | Palesa N

Point of Sale

Stock Check

Sales history

Check medicine

Change tax category

ID	Name	Type	Category	Price	Available	Expiry
1	Paracetamol 500mg	Tablet	Prescribed	35.5	100	2027-06-30
2	Cough Syrup	Syrup	Over the Counter	48.0	35	2026-11-30
3	Vitamin C	Tablet	Over the Counter	62.75	8	2026-09-15
4	Panado	Tablets	Over the Counter	35.5	95	2027-06-30
5	Panado	Syrup	Over the Counter	42.0	80	2027-06-30
6	Adcodol	Tablets	Over the Counter	38.0	75	2027-06-30
7	Myprodol	Capsules		65		2027-06-30
8	Allergex	Tablets		70		2027-06-30
9	Allergex	Syrup		60		2027-06-30
10	Compral	Tablets		55		2027-06-30
11	Corenza C	Tablets	Over the Counter	48.0	60	2027-06-30
12	Sinutab	Tablets	Over the Counter	52.0	51	2027-06-30
13	Adco-Dol	Tablets	Prescribed	40.0	75	2027-06-30
14	Adco-Salbutamol	Tablets	Over the Counter	46.0	50	2027-06-30
15	Adco-Zolpidem	Tablets	Over the Counter	68.0	42	2027-06-30
16	Adco-Naproxen	Tablets	Over the Counter	44.0	60	2027-06-30
17	Benlyn	Syrup	Over the Counter	65.0	38	2027-06-30
18	Lindagon	Capsules	Over the Counter	58.0	50	2027-06-30
19	Lindagon	Syrup	Over the Counter	62.0	45	2027-06-30
20	Betadine	Solution	Over the Counter	72.0	35	2027-06-30
21	Strepsils	Lozenges	Over the Counter	39.0	84	2027-06-30

Tax category

Select medicine category

Prescribed

Prescribed

Over the Counter

Generated Bill window

The bill shows the pharmacy details, customer name, staff member, products purchased, subtotal, tax, total, payment method, tendered amount and change.

 Customer Bill ✕

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HEALTHFIRST PHARMACY
12 Main Street, Johannesburg
Tel: 011 555 0100 | VAT registered

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Customer: Christopher Van der Merve
Served by: Palesa N
Sale number: 13
Date: 05 Sep 2026, 06:48 AM

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Items purchased

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12 Sinutab | Over the Counter | Qty: 1 | Tax: R7.80 | Total: R59.80
75 Adco-Naproxen | Over the Counter | Qty: 3 | Tax: R19.80 | Total: R151.80
42 Amoxil | Over the Counter | Qty: 2 | Tax: R24.60 | Total: R188.60
1 Paracetamol 500mg | Prescribed | Qty: 2 | Tax: R0.00 | Total: R71.00
13 Adco-Dol | Prescribed | Qty: 2 | Tax: R0.00 | Total: R80.00

=====

Subtotal: R499.00
Tax: R52.20
Total: R551.20
Payment method: Cash
Tendered: R600.00
Cash change: R48.80

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Thank you for shopping with HealthFirst.

Save to history

Print bill

Close

Each of the four reports

Sales: This report shows sales totals, number of sales and average transaction values within a selected date range.

HealthFirst Pharmacy Inventory System			
Administrator workspace Riyadh K Lassissi			
Admin home Point of Sale Medicines Suppliers Users Sales history Reports			
Sign out			
Report type From date yyyy mm dd To date yyyy mm dd			
Sales report 2026-01-01 2026-12-31 Generate report Export report			
DATE(s.sale_date)	COUNT(*)	SUM(s.total_amount)	AVG(s.total_amount)
2026-09-05	4	1800.10	450.025000
2026-09-04	1	201.83	201.830000
Sales report Records: 2 Sales total: R2001.93			

Item Wise: This report shows the medicines sold, their categories, quantities sold and sales value.

HealthFirst Pharmacy Inventory System

Administrator workspace | Riyadh K Lassissi

Sign out

Admin homePoint of SaleMedicinesSuppliersUsersSales historyReports

Report type

From date yyyy mm dd

To date yyyy mm dd

Item wise report

2026-01-01

2026-12-31

Generate report

Export report

name	medicine_type	medicine_category	SUM(i.quantity_sold)	SUM(i.quantity_sold * i.price_at_sale)
Adco-Naproxen	Tablets	Over the Counter	6	264.00
Panado	Tablets	Over the Counter	3	106.50
Disprin	Tablets	Over the Counter	3	87.00
Sinutab	Tablets	Over the Counter	2	104.00
Sudocrem	Cream	Over the Counter	2	184.00
Buscopan	Injection	Over the Counter	2	220.00
Benlylin	Syrup	Over the Counter	2	130.00
Amoxil	Capsules	Over the Counter	2	164.00
Paracetamol 500mg	Tablet	Prescribed	2	71.00
Adco-Dol	Tablets	Prescribed	2	80.00
Gaviscon	Tablets	Over the Counter	1	69.00
Bonjela	Gel	Over the Counter	1	57.00
Allergex	Syrup	Over the Counter	1	49.00
Vicks	Lozenges	Over the Counter	1	43.00
Otrivin	Nasal Spray	Over the Counter	1	63.00
Gaviscon	Tablets	Over the Counter	1	69.00

Item wise report | Records: 16

Low Stock: This report identifies medicines that have reached or fallen below their reorder level.

HealthFirst Pharmacy Inventory System

Administrator workspace | Riyadh K Lassissi

Sign out

Admin homePoint of SaleMedicinesSuppliersUsersSales historyReports

Report type

From date yyyy mm dd

To date yyyy mm dd

Low stock report

2026-01-01

2026-12-31

Generate report

Export report

medicine_id	name	medicine_category	quantity_in_stock	reorder_level	(reorder_level - quantity_in_stock)
3	Vitamin C	Over the Counter	8	15	7
62	Vitamin C	Over the Counter	8	15	7

Low stock report | Records: 2

Expiry: This report shows medicines that are expiring within the next month, allowing the pharmacy to manage stock before expiry.

HealthFirst Pharmacy Inventory System

Administrator workspace | Riyadh K Lassissi

Sign out

Admin homePoint of SaleMedicinesSuppliersUsersSales historyReports

Report type

From date yyyy mm dd

To date yyyy mm dd

Expiry report

2026-01-01

2026-12-31

Generate report

Export report

medicine_id	name	medicine_category	expiry_date	DATEDIFF(expiry_date, CURDATE())
3	Vitamin C	Over the Counter	2026-09-15	10
62	Vitamin C	Over the Counter	2026-09-15	10

Expiry report | Records: 2

Please refer to the read me file for additional information based on the project.